Name: Date: Period:

Car Lab #1

Purpose

To determine the ways in which a position graph represents an object's motion.

Materials

- Blue car and Red car
- Dry erase marker and an eraser
- Meter stick
- Timer

Procedure

Before you get started, measure the time it takes each car to travel 1.00 meter. Record the time and calculate the velocity of each car using the equation $v = \Delta x / \Delta t$.

Red Car	
Time to travel 1.00 m (s)	Velocity (m/s)

Blue Car	
Time to travel 1.00 m (s)	Velocity (m/s)

When you are ready to start the experiment:

1. Make a "track" that is 5 meters long. Draw a line and label at "0 m," "1 m," "2 m," "3 m," "4 m," and "5 m."
2. Put the car at 0 m. Count down, release the car, and have the timer start calling out the time every 2 seconds.
3. Every 2 seconds for 12 seconds, make a mark at the location of the car.
4. Once the car has finished its trip, measure each 2-second mark relative to the zero point.
5. Repeat 3 times
6. Repeat for the other car

Data

Experiment #1: Red Car				
Time (s)	Position (m)			
	Trial #1	Trial #2	Trial #3	Average
2				
4				
6				
8				
10				
12				

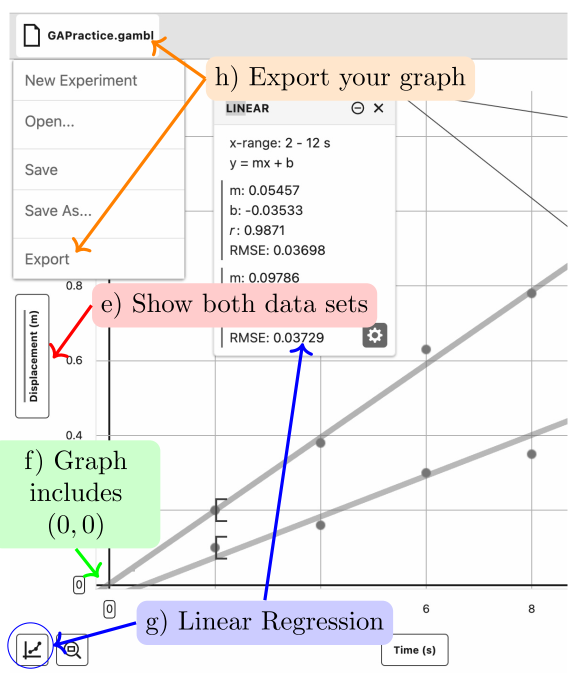
Name: Date: Period:

Experiment #2: Blue Car				
Time (s)	Position (m)			
	Trial #1	Trial #2	Trial #3	Average
2				
4				
6				
8				
10				
12				

How to Make a Graph on *Graphical Analysis*

Open up *Vernier Graphical Analysis* on your laptop.

a) When the program opens, choose "Manual Entry."
b) Click the three dots next to the "X" and "Y" columns in Data Set 1. Click "Column Options" Label the Name an Units for each variable.
c) To enter the second data set, click the three dots next to Data Set 1 and choose "Add new Data Set."
d) Enter your data in each Data Set
e) To display the data from both Data Sets, you will need to click on the y-axis and click the switches next to both data sets
f) **Important!** Make sure that your graph includes (0,0). You can go to "Edit Graph options in the" menu to change your domain and range.
g) Apply a linar curve fit using the button bottom left corner of the screen.
h) Export the graph as a `.png` file using the menu at the top right corner of the screen (Click the file name). Save this file and upload it to Schoology



Analysis

Make a graph on *Graphical Analysis* of your two data sets **Time be x, Average displacement be y.** Plot a best fit line for each data set. Write down the equation for each of the best fit lines below. You should also export the graph and upload it to Schoology

Equation for Red Car:

Equation for Blue Car:

Write a paragraph explaining what you did on this part of the lab. Your paragraph should address: (1) what you graphed and what the graph looked like, (2) why the shape of the graph makes sense, and (3) what the physical meaning is of the *slope* and the *y-intercept*.

Conclusion

You calculated the velocity of each car on the first page. Compare the velocities with the slopes you calculated for each car by calculating a percent error.

$$\text{Percent Error} = \frac{|\text{measured} - \text{expected}|}{\text{expected}} \times 100\%$$

Calculate the percent error and then *write a paragraph commenting on your percent error.* Your paragraph should address: (1) How reasonable is your percent error? (2) Identify three to four factors that could have created uncertainties in your measurements and explain how each factor may have influenced your percent error. (3) What can you say you learned about the position graphs of objects and how they are related to the motion of those objects?

Percent Error (Red Car): ____________________ Percent Error (Blue Car): ____________________